

Question			Marking details	Marks available					Prac
				AO1	AO2	AO3	Total	Maths	
	(b)	(i)	Force in direction of travel required / $280 \cos 35^\circ$ needed Accept: force and distance not in the same direction Accept: horizontal component needs to be calculated or used Don't accept not all the 280 N is used in pulling the sled or pulled at an angle	1			1		
		(ii)	$280 \cos 35^\circ$ or 229.4 N (1) Substitution: $W = 229.4$ (ecf) $\times 3\,000$ (1) $[W = 6.88 \times 10^5 \text{ J}]$ Substitution: $P = \frac{6.88 \times 10^5}{1200}$ ecf on W and t (1) $P = 573.4 \text{ W}$ unit mark or suitable unit alternative (1) Alternative: $280 \cos 35^\circ$ or 229.4 N (1) Calculation of $v = 2.5 \text{ m s}^{-1}$ (1) Substitution: $P = Fv$ i.e. $P = 229.4$ (ecf) $\times 2.5$ (1) $P = 573.4 \text{ W}$ unit mark or suitable unit alternative (1)	1	1 1 1		4	4	
			Question 5 total	4	7	0	11	4	0